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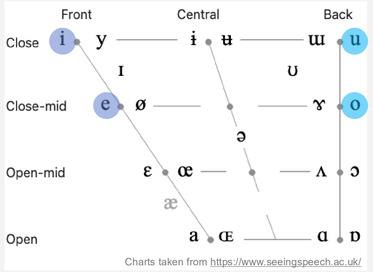
Do six-month-olds use vowel harmony to segment speech?

Helen Shiyang Lu[a], Alexis K. Black[a], & Janet F. Werker[b]

[a] School of Audiology and Speech Sciences and [b] Department of Psychology, University of British Columbia

Introduction

- **Vowel harmony (VH)** is a phonological process in which vowels within a word agree on certain phonetic feature[1]
  - e.g., backness in Turkish
    çocuk-lar "children"
    ev-ler "houses"



Charts taken from https://www.seeingspeech.ac.uk/

- Disharmonies can signal word boundaries, making VH a potential cue for word segmentation[2]
- Infants learning VH languages (e.g., Turkish[3] and Akan[4]) distinguish **harmonic** vs. **disharmonic** words by 6 months, and use VH for word segmentation by 9-11 months[5]
- Recent evidence suggests an **early, broad sensitivity to VH**:
  - 4-month-old infants primarily exposed to non-VH languages prefer **harmonic** compared to **disharmonic** sequences[6]
  - 7-month-old English-learning infants (a non-VH language) can segment speech using VH cues[7]
- This sensitivity seems to **attenuate without relevant input**:
  - By 7-8 months, infants exposed primarily to non-VH language(s) no longer prefer **harmonic** words[6,7], a pattern consistent with perceptual attunement
- Bilingual infants may follow a different developmental path[8]:
  - Experience with multiple languages could maintain broader perceptual sensitivity[9]
  - However, exposure to more than one non-VH languages may also accelerate the down-weighting of VH as a cue

Does vowel harmony sensitivity persist at 6 months without relevant input, and does this differ between monolingual and bilingual infants?

Participants

- Seventy 6-month-old infants with no VH language exposure
  - 38 monolinguals: >= 90% English
  - 32 bilinguals: >= 25% English & >= 20% non-VH language

Methods – Central Fixation

Baseline Phase (testing preference)

- 4 fixed-length trials with a disyllabic word repeated 16 times, with 500-ms interstimulus interval (ISI)

ke-mi / *no-gu* **vs.** *ki-mo* / *nu-ge*

Familiarization Phase

- 108-s continuous speech stream (45 repetitions) with no acoustic, prosodic, or statistical cues to word boundaries

do-tu-be-pi-to-pu-de-bi

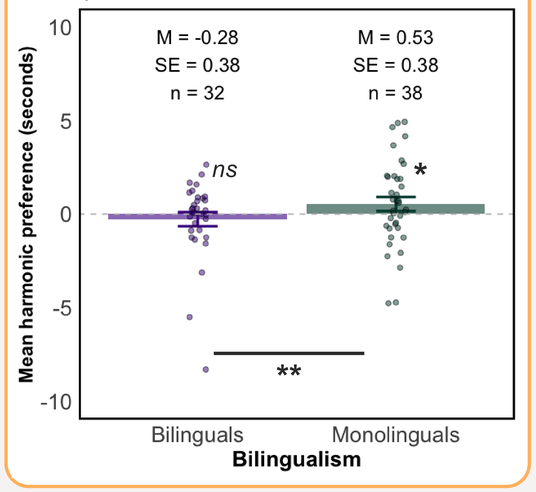
Test Phase (testing segmentation)

- 8 infant-controlled trials presenting a disyllabic sequence from the familiarization stream with 500-ms ISI

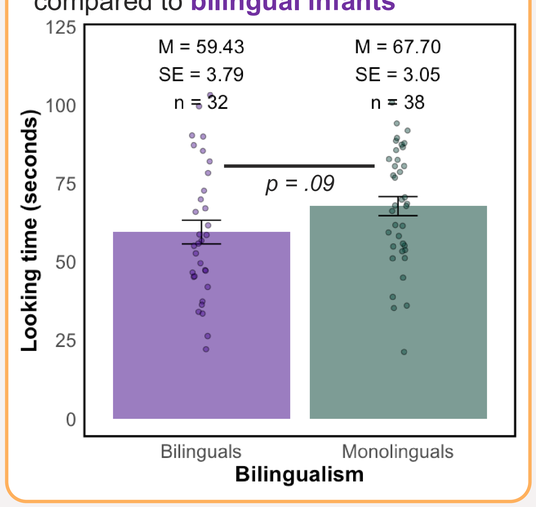
to-pu / *de-bi* **vs.** *bi-do* / *tu-be*

Results and Discussion

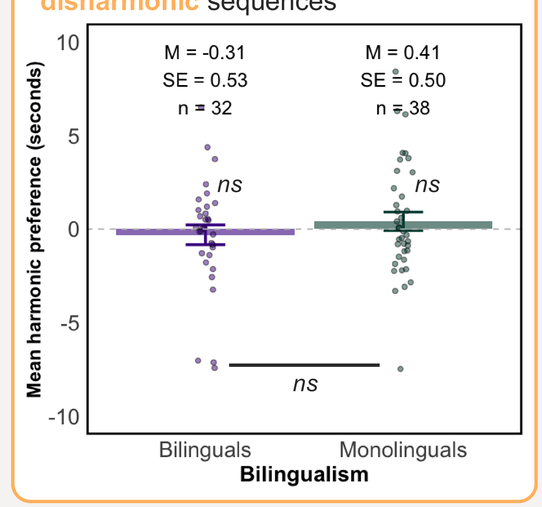
At baseline, **monolingual infants** preferred listening to **harmonic** compared to disharmonic words



In familiarization, **monolingual infants** had *marginally longer* looking times compared to **bilingual infants**



At test, neither group showed differentiation between **harmonic** and **disharmonic** sequences



Preference for Vowel Harmony

- Without relevant input, **English monolingual infants**, but not bilingual infants, prefer **harmonic** over disharmonic sequences at 6 months.
- Perhaps English contain more harmonic forms than expected:
  - Preliminary analyses of the BabyView dataset[10]: ~70% of disyllabic words in English agree in vowel rounding (both rounded or both unrounded), but also ~70% for syllables across word boundaries
- Exposure to more than one non-VH languages may have accelerated the decrease in VH sensitivity

Using VH Cues for Segmentation

- At 6 months, we **did *not* find evidence** of either English monolingual infants or bilingual infants using VH as potential cues for word segmentation
- Potential reasons for different findings from prior work[7]:
  - Central fixation vs. Head-turn preference
  - Slower (~3Hz) vs. Faster (~7Hz) syllable rate

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